The Index Effect in Sweden

A Replication and Extension of
Greenwood and Sammon (2025)

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Bachelor thesis research draft
Stockholm School of Economics
BE451 Degree Project in Finance
Autumn 2026

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Examiner: Riccardo Sabbatucci

Revised 12 November 2026. This is AI-generated research text for author review, not a submission version. The course prohibits AI directly writing the submitted thesis. Appendix G gives the disclosure and verification status. Final submission requires author-written prose and the official cover template with verified student details.

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Abstract

We replicate and extend Greenwood and Sammon's (2025) study of the index effect using OMX Stockholm Benchmark changes from July 2007 through June 2026. The complete sample contains 256 additions and 215 deletions. Addition returns from before announcement through after implementation fall from 8.52% in 2007–2011 to 1.50% in 2022–2026, but do not decline monotonically. Across 2017–2026, the average remains 2.15%, with a 95% interval of [1.01%, 3.39%]. An exploratory timing comparison shows that the median announcement-to-implementation gap falls from fourteen to six trading days for additions. On identical event cohorts, returns over the common initial seven trading days are 1.19% and 1.80%; the full-window decline is concentrated in the remaining interval. This accounting does not establish that calendar changes cause the decline. Reported fund purchases and concentrated implementation trading provide evidence of benchmark-related activity, while changing holdings coverage limits historical multiplier comparisons. The results distinguish a positive later addition association from the less precisely resolved latest period. They establish neither economic disappearance nor a permanent causal price effect of index membership.

Keywords: index effect; replication; passive investing; reconstitution; fund holdings.

1 Introduction

Why have stock-price reactions to index changes weakened as index investing has grown? Greenwood and Sammon (2025) examine this question for the S&P 500 and document a substantial decline in the returns associated with additions and deletions. Their published abstract reports average addition returns below 1% and deletion returns of 0.1% in their latest decade. This finding matters because an index change alters the portfolio that benchmark-tracking investors seek to hold. A smaller price reaction can therefore reflect changes in how markets accommodate predictable trading, rather than a smaller role for index investing itself. Distinguishing these possibilities requires examining trading alongside returns.

This paper replicates and extends the empirical approach of Greenwood and Sammon in the Swedish equity market. Our research question is: *How has the return pattern around OMX Stockholm Benchmark reconstitutions changed, and how does that pattern relate to measured benchmark-linked trading?* We first examine the evolution of announcement, implementation and total-window returns. We then consider announcement timing, the characteristics of affected securities, reported fund holdings and the concentration of trading around implementation. This sequence follows the economic question of the original study while adapting the measures to the Swedish index and available data.

The Swedish setting is informative for two reasons. First, OMX Stockholm Benchmark, hereafter OMXSB, selects constituents with reference to trading activity and industry representation; membership does not follow a single market-capitalization cutoff (NASDAQ OMX, 2013; Nasdaq, 2026). Its scheduled reviews allow us to distinguish the announcement of a change from its implementation. Second, fund-level holdings allow us to examine the quantity of measured benchmark-linked trading alongside the return response. We supplement the commercial holdings data with reports from Finansinspektionen, the Swedish Financial Supervisory Authority. These features permit an extension beyond a comparison of national average returns: we can examine whether price and trading patterns tell a consistent story within the Swedish setting.

Our event inventory covers 39 reviews from July 2007 through June 2026. The main return sample contains 256 additions and 215 deletions. For each event, we calculate market-adjusted returns relative to the OMXSB gross total-return index. The main window begins one trading day before the announcement and ends one trading day after the change becomes effective. We report four historical blocks, together with comparisons between them and pooled estimates for the later decade. Uncertainty estimates account for the fact that several securities enter or leave at the same review. The holdings analysis uses its own matched sample because historical holdings are not available for every return observation.

Three findings emerge. First, the Swedish return pattern shows lower recent averages than in the earliest years, but no smooth disappearance. Average addition returns are 8.52%, 0.25%, 2.85% and 1.50% in 2007–2011, 2012–2016, 2017–2021 and 2022–2026, respectively. Pooling 2017–2026 gives an average of 2.15%, with a 95% confidence interval of [1.01%, 3.39%]. For 2022–2026 alone, the interval is [-0.10%, 3.22%]. The latest observations therefore leave substantial uncertainty about the remaining return, rather than

establishing that it is economically negligible. Deletion returns are less precisely estimated: the later-decade mean is -1.04%, with an interval spanning zero.

Second, the time between announcement and implementation has changed. Across the reviews contributing additions, the median gap falls from fourteen trading days in 2007–2011 to six in 2022–2026. The full event window consequently covers different durations. In an exploratory timing extension, we compare a common seven-trading-day window, from the day before announcement through five trading days after it. This window ends before implementation for every event, and each period retains exactly the same stocks as its main return estimate. Average addition returns over these seven days are 1.19% in the earliest period and 1.80% in the latest, compared with 8.52% and 1.50% over the full windows. The large early-to-late difference is therefore concentrated after the common initial window. This finding makes the time available for returns to accumulate an important part of the historical comparison.

Third, the return response coincides with measurable benchmark-linked trading. Among 136 additions with matched holdings data in 2017–2026, the mean event-window return is 2.71% and measured net purchases represent 1.03% of shares outstanding. The ratio of those means is 2.63 percentage points of return per percentage point of measured ownership change. The Finansinspektionen comparison provides a check on the scale of the holdings measure: in a matched sample of 106 later additions, the reported ownership changes are 1.14% and 1.05% under the two reporting sources. Historical comparisons are less secure because observed investor coverage changes and the earliest holdings samples are small.

Trading also becomes more concentrated immediately before implementation. For additions, the final trading day before the effective date accounts for 9.44% of the surrounding 45-day trading volume in the earliest period and 15.24% in the latest. Deletions display a similar increase. Relatively small recent implementation-window returns therefore coexist with concentrated trading around the date on which benchmark portfolios change. The amount and timing of trading need not move proportionally with the contemporaneous return. Daily volume identifies this concentration of activity without distinguishing closing-auction trades from other transactions.

The timing extension also follows a fixed event sample through twenty trading days after implementation. For additions in 2017–2026, the cumulative market-adjusted return over this longer window is 3.30%, with a 95% interval of [1.31%, 5.28%]. The positive result remains under separate exclusions of an influential event and simultaneous MSCI changes. The latest-period estimate is more sensitive to those exclusions. Together, the common initial window and the longer horizon show why the timing of the measured return matters for interpreting the headline averages.

Our contribution is a Swedish replication that connects the return comparison to announcement timing, reported holdings and implementation trading. A positive addition return remains visible over the later decade, while the shorter recent announcement-to-implementation gap changes the interpretation of the historical comparison. These findings are observational: they do not establish that the calendar caused the change in returns, that the initial-window means are equivalent, or that index demand causes a permanent price change. Selection, firm news and changing holdings coverage remain possible influences. The return-to-holdings ratio is consequently descriptive rather than an identified demand

elasticity. The economic contribution is evidence on which return and trading patterns extend to Sweden and how their measurement affects the answer.

Section 2 positions the study in the related literature. Section 3 describes the index, data and empirical methods. Section 4 presents the replication results, followed by their interpretation and the extension in Section 5. Section 6 concludes.

2 Related literature

Our closest reference is Greenwood and Sammon (2025), who study the declining S&P 500 index effect and examine whether changes in demand, the composition of index events and trading conditions account for it. We apply these empirical questions to OMXSB reconstitutions, adapting the benchmark, sample period and holdings measures to Swedish data. Our timing extension examines how the historical comparison changes when the announcement window has a fixed duration and uses the same stocks as the full event window. We also compare holdings measurements across two reporting sources and follow cumulative returns on a fixed event sample after implementation. These exercises connect the return comparison to its institutional timing and measured trading counterpart.

The interpretation of index-event returns builds on Shleifer (1986) and Harris and Gurel (1986). The former connects index additions to downward-sloping demand for individual stocks, while the latter examines price pressure and subsequent reversal. These studies motivate looking beyond the announcement return to the timing of trading and the subsequent price path. A positive cumulative return at a later date is relevant to that comparison, but does not by itself distinguish a lasting demand response from information or selection associated with index membership.

Chang et al. (2015) study the price effects of indexing through a regression discontinuity design. Their setting illustrates the additional identifying structure needed to interpret an index comparison causally. OMXSB's selection criteria differ, and the present study does not reconstruct an equivalent assignment boundary. Our composition adjustments and comparison securities therefore describe conditional return differences rather than treatment effects.

Finally, Chinco and Sammon (2024) show why the investor population relevant to passive ownership can be broader than explicitly labeled index funds. This distinction matters for measuring the demand associated with OMXSB changes, where benchmark variants and benchmark-linked mixed funds can hold the same securities. We consequently distinguish observed changes in the covered funds' positions from total passive demand. The combined contribution is evidence on the return pattern and its trading counterpart in an additional institutional setting, with a direct measurement comparison for reported holdings.

3 Data and methodology

3.1 The Swedish setting

OMX Stockholm Benchmark (OMXSB) provides a setting for examining the question in Greenwood and Sammon (2025): how do returns around index changes evolve alongside the associated trading? The index represents a selection of large and actively traded Stockholm-listed securities. Selection combines liquidity and industry representation, rather than a single market-capitalization cutoff. The current rules use turnover eligibility and free-float capitalization within industry supersectors; the index is weighted by free-float capitalization (Nasdaq, 2026). A size screen alone therefore does not reproduce the assignment rule.

The distinction between announcement and implementation is central. The announcement identifies the revised portfolio; implementation is when it becomes effective. We call their recorded trading dates AD and ED. When effectiveness is at the opening of ED, ED-1 is the last trading day under the old composition and a natural day for portfolio adjustment. Small returns around ED can therefore coexist with substantial trading on ED-1 if prices have already adjusted.

Several securities enter or leave at each scheduled review. Historical dates come from the individual notices. Early observations include January and July implementations; later ones mainly use June and December. The current rules announce changes after the close on the sixth trading day before effectiveness. That calendar is not imposed retrospectively on earlier events (NASDAQ OMX, 2013; Nasdaq, 2026).

The announcement-to-implementation interval was generally longer early in the sample than recently. A return between those dates consequently covers different numbers of trading days over time. We report that interval and supplement the full event window with a fixed seven-day announcement window. The full window measures the whole episode; the fixed window compares the same amount of time immediately around the news.

Selection rules also change. The May 2012 notice introduced revised industry-diversification criteria and announced 11 additions and 24 removals (NASDAQ OMX, 2012). That large review motivates checks of review influence. It is not treated as a natural experiment, because its effect is not isolated from contemporaneous conditions.

3.2 Sample and sources

The event inventory contains 536 changes: 284 additions and 252 deletions across 39 reviews, with effective dates from July 2007 through June 2026. Each event has a security identifier, announcement date, effective date and source reference. LSEG supplies security returns, prices, volume, capitalization, shares outstanding and closing quotes. The market benchmark is the OMXSB gross total-return index, OMXSBGI. Historical identifier matching and price availability reduce the imported return panel to 487 candidates.

Ten deletions flagged for offers or delisting-related circumstances are excluded. Six further candidates lack a complete main return window. The final sample contains 256 additions and 215 deletions. Each type appears in 38 reviews; their union covers all 39.

Table 1: Construction of the return sample

Stage	Additions	Deletions
Documented membership changes	284	252
Imported return candidates	261	226
After deletion exclusions	261	216
Complete main return window	256	215

The source list contains 536 membership changes in 39 review cycles from 2007 through the first review of 2026. The imported return panel contains 487 candidates. Ten deletions flagged for offers or delisting are excluded; six remaining candidates lack a complete main window. The final sample contains 471 event observations. Each event type spans 38 review cycles; their union spans 39. Source: Authors' calculations; exact inputs are listed in the replication package.

Missing historical observations may differ from observed securities, so availability is not assumed to be random.

We report four calendar blocks: 2007–2011, 2012–2016, 2017–2021 and 2022–2026. The first and last calendar years are incomplete. We also pool 2007–2016 and 2017–2026 to distinguish a broad later-period pattern from the less precise latest block. Granular and pooled results are both retained. These divisions and the supplementary analyses are retrospective, not registered before the original results were known.

Table 2: Coverage across periods

Type	Period	Source events	Return sample	Holdings sample	Reviews
Addition	2007–2011	48	33	8	9
Addition	2012–2016	58	54	27	10
Addition	2017–2021	90	82	58	10
Addition	2022–2026	88	87	78	9
Deletion	2007–2011	54	33	7	9
Deletion	2012–2016	53	46	25	10
Deletion	2017–2021	66	61	40	10
Deletion	2022–2026	79	75	68	9

The source inventory precedes security matching, exclusions and return requirements. Return sample requires complete AD-1 through ED+1 returns after the stated deletion exclusions. Holdings sample additionally requires the broad demand measure and retention under the share-adjustment screen, with a nonmissing matching return. Reviews counts distinct review cycles in the return sample. The figures describe coverage, not random attrition.

Table 2 shows why coverage matters for comparisons over time. The return and holdings columns answer different questions and should not be combined as if their populations were identical. A smaller holdings sample can change its mean return as well as its measured purchases. We therefore show counts with the component means whenever demand is used, and retain the full available return sample for the baseline event study. The appendix traces the especially restrictive early adjustment screen.

Security returns use the vendor's total-return field where available and a closing-price return fallback otherwise. The fallback affects seven imported events and 104 event-day

observations in the main windows. Removing implicit price forward filling leaves the main CARs unchanged. This arithmetic agreement does not authenticate every historical corporate action or dividend. Appendix B records the remaining source checks.

Fund positions come from LSEG/Lipper extracts and a dated list of benchmark mandates. The main population includes strict OMXSB followers, related benchmark variants, and the OMXSB-linked component of mixed funds. We call it the broad benchmark-linked group. Narrower definitions accompany it. A separate Finansinspektionen (FI) reporting channel permits an overlapping-sample check for 2019–2026. Neither source is a comprehensive record of all passive capital or implementation-day orders.

Closing bid and ask quotes and daily volume describe trading conditions. The quoted spread is the bid–ask difference divided by the midpoint; the annual series uses capitalization weights among index members. Daily volume concentration divides each day's volume by the event's total over ED-22 through ED+22, holding the denominator at 45 trading days. These variables do not directly measure closing-auction activity or the execution cost of a large order.

3.3 What is replicated and what is adapted

The analysis follows Greenwood and Sammon's economic sequence: document returns, examine changing firm characteristics, consider overlapping index activity and anticipation, and relate returns to measured index demand and trading conditions. Table 3 states the correspondence. This is a replication of their empirical approach in a new setting, with explicit adaptations, rather than a reproduction of their US estimates.

Table 3: From Greenwood–Sammon to the Swedish implementation

Component	Anchor study	Swedish analysis
Return pattern	S&P 500 additions/deletions; announcement, implementation and full windows	OMXSB; same event-window questions, explicitly defined daily-sum CAR
Time comparison	Long US history and decade means	Four Swedish calendar blocks and two pooled decades; incomplete 2026
Composition	Period indicators and four firm characteristics	Same regression structure; Swedish measurement proxies and common samples
Index demand	Tracking-trade classification; quarterly holdings	Dated mandate groups; available pre/post reports; FI comparison
Other index activity	Migrations within S&P index families	Coincident MSCI changes; a distinct overlap diagnostic
Anticipation	Pre-announcement returns and longer starting window	Pre-return windows and AD-20 to ED+1 ratio; historical eligibility not fully reconstructed
Liquidity	Effective spreads and implementation costs	Closing quoted spreads and daily volume concentration
Extensions	Anchor comparison provides the baseline	Fixed-duration timing, within-firm share classes and separate holdings reporting

The mapping distinguishes conceptual replication from exact numerical reproduction. Detailed source-version references and adaptations appear in Appendix A. Differences in periods, sample selection, holdings dates and return conventions prevent interpreting raw cross-country differences as causal effects.

Three differences are especially consequential. First, the Swedish return sample is larger than the holdings sample. Every demand-scaled estimate therefore uses identical events in numerator and denominator. Second, the production fund classification uses dated mandates, whereas Greenwood and Sammon identify trackers by their propensity to undertake index-tracking trades. The trading-based Swedish classification is a separate diagnostic. Third, coincident MSCI changes are not equivalent to migration between S&P index families. Excluding them tests whether the Swedish return pattern is also present without that overlap; it does not reproduce the US migration decomposition.

Appendix A records the detailed correspondence and source versions.[1]

3.4 Returns and changes over time

The Swedish primary measure sums daily stock returns in excess of the benchmark:

$$CAR_i[a, b] = \sum_{t=a}^{b} (r_{it} - r_{mt}), \tag{1}$$

where r_{it} is the stock's daily return and r_{mt} is the OMXSBGI return, both in percent. The full window includes daily returns from AD-1 through ED+1. Announcement

[1] Published US headline estimates are taken from the 2025 abstract. Available full author manuscripts contain some different numerical results. Original US code was not recovered, so implementation equivalence is not asserted where the return conventions remain ambiguous.

and implementation windows include AD-1 through AD+1 and ED-1 through ED+1, respectively. Endpoints are trading days; every required daily return must be available.

Greenwood and Sammon express abnormal performance as cumulative stock return minus cumulative market return. The Swedish implementation uses the arithmetic daily sum in Equation (1). We also compute compounded stock return minus compounded index return over identical included days, and a closing-price-endpoint alternative. These are definition checks rather than silent changes to the primary measure. Announcement and implementation means should not be added: the full window includes intervening days, and completeness requirements can select different observations.

Period means are estimated separately for additions and deletions. To examine whether observable firm characteristics explain their differences, we follow the structure of the anchor's composition regression:

$$CAR_i = \sum_p \alpha_p \mathbf{1}\{i \in p\} + \boldsymbol{\beta}'(X_i - \overline{X}) + \varepsilon_i. \tag{2}$$

Period indicators replace the intercept. X_i contains abnormal turnover, relative size, residual return variance and analyst coverage. Centering makes the period coefficients interpretable at the common-sample average characteristics. We show unadjusted and adjusted estimates for the same events, separating adjustment from the loss of observations with missing controls.

The Swedish controls approximate the US variables. Turnover is measured before announcement, relative capitalization at AD-1, residual variance over an earlier return history, and coverage from the preceding month's forecasts. A forecast count need not equal the number of unique analysts at the last earnings announcement. Capitalization conventions can differ across share classes. These controls describe composition; they do not reproduce every selection criterion or establish causal assignment.

Direct contrasts compare the first and last blocks, the two latest blocks and the two pooled decades. A separate linear annual trend checks whether the result depends on period boundaries. A significant mean in one period and an insignificant mean in another do not themselves establish a difference. The fixed announcement window, AD-1 through AD+5, contains seven daily returns and ends before ED for every observed event. It was selected after inspecting timing metadata and is a supplementary diagnostic, not a replacement chosen for its significance.

3.5 Fund demand and the multiplier

Greenwood and Sammon relate returns to the fraction of shares bought or sold by trackers. Their multiplier M expresses the return associated with one percentage point of measured demand. We retain these economic units. Measured demand for an addition is covered funds' net purchases divided by shares outstanding. For deletions, sales and negative returns are sign-adjusted so that index-directed movements have positive signs:

$$D_i = 100 s_i \frac{\sum_f \Delta H_{if}}{S_i}, \qquad M_p = \frac{\overline{s_i CAR_{ip}}}{\overline{D}_p}, \tag{3}$$

where s_i is $+1$ for additions and -1 for deletions, ΔH_{if} is the change in shares reported by fund f, and S_i is shares outstanding. For example, a mean return of 2.71% divided by measured purchases of 1.03% gives approximately 2.63. This is a ratio of means, not an average of event-specific ratios.

The holdings calculation uses a pre-announcement report within 120 calendar days and a post-event report no later than 45 days after ED. Missing security positions can be assigned zero when the fund is otherwise classified as reporting. A security extract alone cannot verify that every absent position is genuinely zero. An adjustment-factor screen and the available shares-outstanding denominator further restrict the sample. Appendix C states their exact operation. The screen is not simply an exclusion of splits occurring during the event window.

Coverage affects interpretation. Observing more funds can raise measured purchases without a commensurate increase in total demand. A small denominator can also make M large and imprecise. We report the component means and event counts, compare fund definitions and use FI as a separate reporting check. The ratio remains descriptive; we do not convert it into an identified demand elasticity. Its uncertainty accounts jointly for the numerator and denominator, with technical details in the appendix.

3.6 Uncertainty

Securities in a review share dates and market conditions. Main return inference therefore clusters by review, an adaptation to a setting with many events but only 39 review cycles. The baseline return table and return-exclusion checks report 95% intervals from a restricted wild cluster bootstrap with 9,999 draws. The latest block has nine clusters per event type, so precision remains limited. Appendix D documents the implementation and dependence sensitivities (Petersen, 2009; MacKinnon et al., 2023).

We interpret magnitudes and intervals before significance. An interval containing both zero and returns above 3% cannot establish economic disappearance. A limited adjustment for ten related return tests and a clearly labeled practical-equivalence check appear in the appendix. They describe uncertainty in an exploratory study rather than create a preregistered design.

4 Empirical results

4.1 Does the disappearing index effect appear in Sweden?

The Swedish data show lower full-window returns than in the earliest period, but not a steady disappearance. Table 4 reports the announcement, implementation and full-window means. Addition CARs are 8.52%, 0.25%, 2.85% and 1.50% across the four successive periods. The large early value is followed by a near-zero mean and then a recovery. Figure 1 displays the estimates with their uncertainty, making that non-monotonic pattern explicit.

Table 4: Market adjusted returns around index changes

Type	Period	N	G	AD	ED	Total	95% interval	p
Addition	2007–2011	33	9	0.96	1.16	8.52	[2.46, 14.74]	0.027
Addition	2012–2016	54	10	1.87	-0.73	0.25	[-3.73, 5.30]	0.908
Addition	2017–2021	82	10	-0.16	0.38	2.85	[1.26, 5.20]	0.005
Addition	2022–2026	87	9	1.20	-0.09	1.50	[-0.10, 3.22]	0.062
Deletion	2007–2011	33	9	-0.77	-0.09	-6.16	[-11.41, -2.10]	0.007
Deletion	2012–2016	46	10	-1.71	-0.85	-2.47	[-14.42, 3.60]	0.522
Deletion	2017–2021	61	10	-2.09	0.89	-0.43	[-3.43, 3.02]	0.759
Deletion	2022–2026	75	9	-0.71	-1.23	-1.53	[-4.24, 0.89]	0.279

Returns are percentages. AD denotes CAR $[AD-1, AD+1]$, ED denotes CAR $[ED-1, ED+1]$, and Total denotes CAR $[AD-1, ED+1]$, where AD is the recorded announcement trading date and ED the effective trading date. The benchmark is OMXSBGI. Means weight events equally. N and G are event and review counts for Total; AD and ED use their own complete cases. Intervals and two-sided p-values for Total use the restricted wild cluster bootstrap with 9,999 draws, Webb weights below 12 clusters and Rademacher weights otherwise. These are pointwise, exploratory intervals. No stars are used. Source: Authors' calculations; exact inputs are listed in the replication package.

Deletions have negative means in all four periods: -6.16%, -2.47%, -0.43% and -1.53%. Their intervals are generally wider than those for additions. In particular, the 2012–2016 deletion estimate is sensitive to the weighting of large reviews. The sign of the averages is consistent with the conventional index-event pattern, but the sample does not precisely resolve the magnitude of recent deletion returns.

The published US finding in Greenwood and Sammon (2025) is a useful benchmark for the question, not a directly comparable control group. Their published abstract reports an average S&P 500 addition return below 1% and deletion return of 0.1% in 2010–2020. Sweden's later interval begins in 2017 and ends in 2026, uses a different benchmark and review institution, and contains different firms. We consequently ask whether a similarly small return is evident within the Swedish series, rather than subtracting the US number and calling the difference a country effect.

For 2017–2026 as a whole, addition CAR averages 2.15%, with a 95% interval of [1.01%, 3.39%], across 169 events and 19 reviews. The corresponding deletion mean is -1.04%, with an interval of [-2.88%, 0.90%]. These are economically different levels of precision.

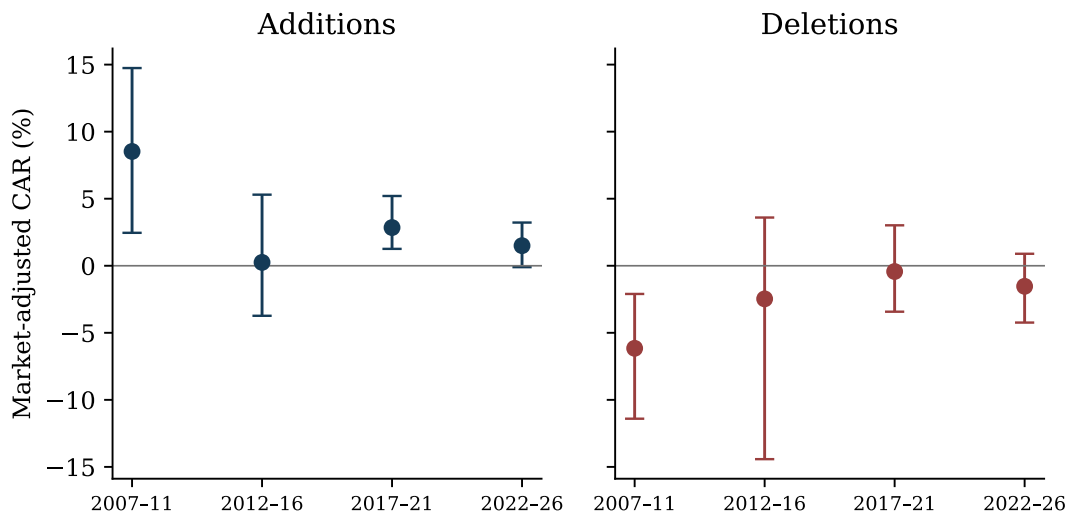


Figure 1: Returns around OMXSB changes by period. Points are event-weighted mean CARs from AD-1 through ED+1; bars are the 95% review-cluster wild-bootstrap intervals reported in Table 4. The vertical scale is identical for additions and deletions. Periods begin in July 2007 and end in June 2026. The estimates describe observed index events, not a fitted trend.

The latest addition estimate, 1.50%, has an interval of [-0.10%, 3.22%]. This permits both a near-zero average and a positive return of economically meaningful size. It does not establish disappearance. The latest block contains only nine review clusters, and its mean is sensitive to particular observations and overlapping index events, as shown below.

The direct time contrasts sharpen the answer. Latest minus earliest addition CAR is -7.03 percentage points, whereas latest minus preceding is -1.35 points, later minus earlier decade is -1.24 points, and the annual linear trend is -0.28 points. Only the endpoint contrast has a pointwise bootstrap interval excluding zero; that result does not survive the limited ten-test multiplicity adjustment. Table A6 in the appendix reports all contrasts. The evidence is therefore strongest for a high earliest-period mean and a positive later-decade association, rather than a statistically established smooth decline.

4.2 How much does the changing event calendar matter?

The full-window comparison covers the entire interval between announcement and implementation. In Sweden, that interval changes substantially. Among reviews represented in the complete addition sample, its median is 14, 16, 15 and 6 trading days in the four blocks. Thus, the latest full window is considerably shorter than the typical earlier window. This is a feature of the institution and measurement, not evidence by itself that timing caused any return difference.

We therefore calculate a fixed initial seven-day window, AD-1 through AD+5, on the same events as the full-window means. It contains seven daily returns and ends before implementation for every event. This choice was made after inspecting the event calendar, and the exercise is exploratory. Table 5 reports both windows and the remaining portion of the full window.

Table 5: The full event window and a common initial seven-day window

Type	Period	Events	Gap	Full	Initial seven	Remainder
Addition	2007–2011	33	14	8.52	1.19	7.34
Addition	2012–2016	54	16	0.25	1.26	-1.01
Addition	2017–2021	82	15	2.85	1.23	1.62
Addition	2022–2026	87	6	1.50	1.80	-0.31
Deletion	2007–2011	33	15	-6.16	-3.54	-2.62
Deletion	2012–2016	46	16	-2.47	-0.86	-1.61
Deletion	2017–2021	61	15	-0.43	-1.82	1.39
Deletion	2022–2026	75	6	-1.53	-0.76	-0.77

Returns are percentages. Gap is the median AD-to-ED distance in trading days across distinct reviews represented in the sample. Full includes AD-1 through ED+1; Initial seven includes AD-1 through AD+5; Remainder includes AD+6 through ED+1. Every row uses identical events and the two components sum to Full. The remainder has varying duration. This timing diagnostic is exploratory; uncertainty for period contrasts is reported in Appendix D.

The addition initial seven-day means are 1.19%, 1.26%, 1.23% and 1.80%. The very high 8.52% earliest full-window mean is therefore not a comparably high initial seven-day response: approximately 7.34 percentage points accrue in the remaining interval. In the latest block, the remainder is approximately -0.31 points. These are within-sample accounting comparisons; they do not establish that the initial seven-day population means are equal or that shortening the calendar explains the difference causally.

The distinction matters for the replication's economic interpretation. A declining full-event CAR need not be a declining immediate response to the announcement. It may reflect where returns occur along an episode whose duration also changes. Both parts of the evidence should be visible. The full window remains appropriate for the entire index change, while the fixed window prevents the early large CAR from being described as a large announcement-week reaction without checking its timing.

Return aggregation is a separate issue. On the same 169 later additions, replacing the arithmetic daily sum with compounded stock return minus compounded benchmark return changes the mean from 2.15% to 2.18%. In the latest block it changes 1.50% to 1.45%. The positive pooled later association is not created by this particular aggregation choice. Alternative closing-price endpoints are also shown in Appendix B; the daily return inclusion convention is kept explicit rather than treated as interchangeable with the window label.

4.3 Composition, coincident index changes and anticipation

Following the anchor study, we next ask whether changing observable characteristics account for the time pattern. Table 6 compares unadjusted and adjusted period estimates on identical complete-case samples. For latest-period additions, the estimate is 1.59% without controls and 1.60% with them. For early additions, the corresponding values are 8.97% and 10.12%. These controls do not explain away the difference in this specification.

Table 6: Composition adjustment on a common event sample

Type	Period	Unadjusted	Adjusted	Adjusted CR1 interval
Addition	2007–11	8.97	10.12	[5.09, 15.16]
Addition	2012–16	0.74	0.55	[-3.01, 4.11]
Addition	2017–21	2.64	2.33	[0.38, 4.29]
Addition	2022–26	1.59	1.60	[0.68, 2.53]
Deletion	2007–11	-5.66	-6.23	[-11.03, -1.43]
Deletion	2012–16	-1.37	-1.42	[-7.61, 4.76]
Deletion	2017–21	-0.77	-0.29	[-3.35, 2.77]
Deletion	2022–26	-1.83	-1.95	[-4.02, 0.12]

Period intercepts from CAR $[AD-1, ED+1]$ regressions. Both columns use the same complete-case sample: 237 additions or 185 deletions. Adjusted estimates control for centered abnormal turnover, relative size, residual return variance and analyst coverage. The columns are not a causal decomposition. Intervals are analytical CR1 review-clustered intervals using t_{G-1}; unstable inverted bootstrap intervals from auxiliary sparse specifications are not used here. Source: Authors' calculations; exact inputs are listed in the replication package.

The comparison is conditional on the available turnover, size, residual-variance and analyst-coverage proxies. It does not prove that composition is irrelevant. Missing controls reduce the sample to 237 additions and 185 deletions, and the proxies do not capture all reasons for membership changes. The common-sample unadjusted column is important: comparing an adjusted result with a larger raw sample would otherwise attribute selection from missing controls to the controls themselves.

Coincident MSCI changes matter for magnitudes. Excluding them lowers the later addition mean from 2.15% to 1.46%, with an interval of [0.50%, 2.47%]. The latest mean falls from 1.50% to 0.73%, with an interval including zero. Overlapping events have unusually large returns. This is a Swedish counterpart to asking whether index-specific institutional details change measured demand, but it is not the US migration mechanism: removing an overlap does not identify the incremental causal contribution of either index.

Table 7 also excludes Sivers separately, applies winsorization, and removes observations flagged in a targeted company-news review. The later-decade addition mean remains positive with an interval excluding zero in these individual checks. The latest-block mean is more sensitive. The news screen covers the 24 most extreme events, so its remaining observations cannot be described as comprehensively news-free.

Anticipation is another explanation considered by Greenwood and Sammon. The Swedish additions have positive pre-announcement performance over a long horizon: CAR from AD-100 through AD-1 averages 12.11%, 18.05%, 11.82% and 18.88% across the four periods. Over the shorter AD-20 through AD-1 window, the means are 0.32%, 4.11%, 0.60% and 2.94%. Neither sequence shows a steady rise that mirrors the full-window decline.

These pre-announcement returns may reflect anticipation, but selection on earlier price and trading performance can also generate them. The historical Nasdaq eligibility rule is not reconstructed sufficiently to separate the two. The pre-announcement windows also use their own complete cases and include AD-1; they cannot simply be added to the

Table 7: Sensitivity of addition returns

Period	Specification	N	CAR	95% interval
2017–26	Baseline	169	2.15	[1.01, 3.39]
2017–26	Without MSCI overlap	157	1.46	[0.50, 2.47]
2017–26	Without screened company news	163	1.91	[1.01, 2.82]
2017–26	Winsorized 1st and 99th percentiles	169	1.92	[0.91, 3.05]
2017–26	Without Sivers	168	1.71	[0.57, 2.96]
2022–26	Baseline	87	1.50	[-0.10, 3.22]
2022–26	Without MSCI overlap	82	0.73	[-0.77, 2.32]
2022–26	Without screened company news	84	1.30	[0.12, 2.43]
2022–26	Winsorized 1st and 99th percentiles	87	1.04	[-0.11, 2.25]
2022–26	Without Sivers	86	0.62	[-0.85, 2.23]

Event-weighted CAR $[AD-1, ED+1]$ in percent, with pointwise restricted wild cluster bootstrap intervals by review. Each exclusion is applied separately. The company-news screen examines the 24 most extreme events, rather than the full event universe, and is outcome-informed. MSCI exclusion changes the target population. Winsorization changes the outcome. These are sensitivity descriptions, not independent replications or specification selection rules. Source: Authors' calculations; exact inputs are listed in the replication package.

Table 8: Pre-announcement returns

Type	Period	N_{20}	AD-20 to AD-1	N_{100}	AD-100 to AD-1
Addition	2007–11	36	0.32	31	12.11
Addition	2012–16	54	4.11	51	18.05
Addition	2017–21	83	0.60	80	11.82
Addition	2022–26	84	2.94	81	18.88
Deletion	2007–11	32	-5.56	31	-16.40
Deletion	2012–16	46	-3.70	43	-9.31
Deletion	2017–21	62	0.54	62	-10.26
Deletion	2022–26	74	-0.48	70	-21.80

Arithmetic market-adjusted returns in percent. Each window uses its own complete cases and includes AD-1, which also belongs to the primary event window. The columns cannot be added to the main-window mean. Positive prior performance is consistent with anticipation but also with selection on earlier returns; it does not distinguish these channels.

full-window mean. Appendix A reports the period results and the longer AD-20-to-ED+1 demand-ratio counterpart used in the original study.

Taken together, composition adjustment, overlap exclusions and pre-announcement returns narrow the explanations without selecting one. Observed characteristics do not remove the pattern in the matched sample; overlapping events have higher sample returns, but the mean remains positive when they are excluded; and earlier returns do not establish a progressively stronger anticipation mechanism. These are useful findings even though they do not constitute causal decompositions.

4.4 Do measured fund purchases explain the return pattern?

The next step follows Greenwood and Sammon's link between the amount bought by trackers and the associated price response. Table 9 reports the mean sign-adjusted CAR, measured purchases or sales, and their ratio on exactly the same events. Every figure has a straightforward economic unit: demand is a percentage of shares outstanding, and M is percentage points of return per percentage point of measured ownership change.

Table 9: Observed fund trading and return per unit of measured demand

Type	Period	Events	Return	Demand	M	95% ratio set
Addition	2007–11	8	8.78	0.15	57.87	[-9.55, 471.88]
Addition	2012–16	27	-1.94	0.52	-3.76	[-15.09, 4.24]
Addition	2017–21	58	3.29	1.11	2.97	[-0.14, 6.30]
Addition	2022–26	78	2.28	0.97	2.35	[-0.21, 4.73]
Addition	2017–26	136	2.71	1.03	2.63	[0.91, 4.37]
Deletion	2007–11	7	-0.19	0.26	-0.72	All real values
Deletion	2012–16	25	2.07	0.48	4.36	[-17.73, 30.17]
Deletion	2017–21	40	0.20	0.99	0.20	[-4.05, 4.89]
Deletion	2022–26	68	1.66	1.18	1.41	[-0.87, 2.83]
Deletion	2017–26	108	1.12	1.11	1.01	[-0.88, 2.61]

Return is sign-adjusted CAR from AD-1 through ED+1, in percent. Demand is the sign-adjusted broad benchmark-linked holdings change as a percentage of shares outstanding. For deletions, negative returns and sales have positive signs. Each ratio divides the two means on identical events retained after the share-adjustment screen. Joint ratio uncertainty is described in Appendix D; it can be unbounded when the denominator is weak. These are descriptive ratios, not identified elasticities.

For additions in 2017–2026, the common holdings sample contains 136 events. Mean CAR is 2.71%, mean measured demand is 1.03% of shares outstanding, and their ratio is 2.63. The return exceeds the all-event mean of 2.15% because the samples differ. The ratio does not imply that an exogenous purchase of 1% causes a 2.63% price rise; it describes the relation between two sample averages under the observed fund coverage.

The early addition ratio, 57.87, is especially fragile. It uses only eight events across five reviews, with measured purchases averaging 0.15% of shares outstanding. Jointly allowing the numerator and denominator to be uncertain gives a 95% ratio set from -9.55 to 471.88. This large number cannot support a precise claim that Swedish market elasticity

has changed by a corresponding factor. The broad later-decade ratio is better resolved, with a set of [0.91, 4.37], while both later five-year ratio sets include zero.

The demand series also needs a coverage interpretation. The broad addition measure is 0.15%, 0.52%, 1.11% and 0.97% across the four blocks. It rises relative to the sparse early observations but does not increase continuously. Reporting coverage and fund mandates evolve, and the share-adjustment screen removes a much larger proportion of early candidates. The observed historical increase therefore combines investor activity with changes in what can be measured.

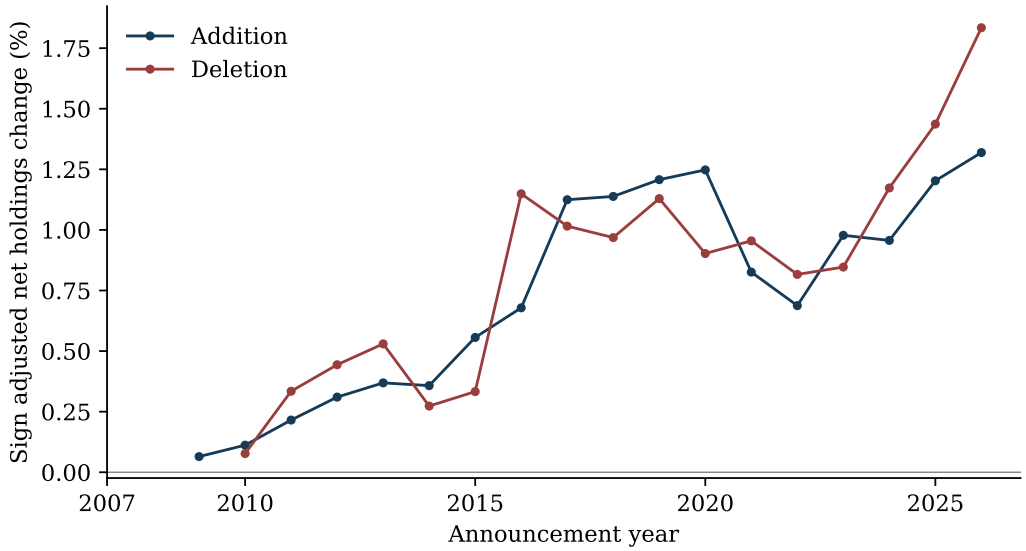


Figure 2: Observed benchmark-linked ownership changes. Annual event means of sign-adjusted net holdings changes as a percentage of shares outstanding. Positive deletion values denote sales. The broad fund definition includes benchmark variants and benchmark-linked mixed-fund holdings. Coverage changes over time; the series is not an estimate of total passive assets or implementation-day orders. The last year is incomplete.

The separate FI reporting channel is informative on a matched recent sample. For 106 additions in 2019–2026, measured demand is 1.14% using FI and 1.05% using the matched Lipper-based measure. Both use the same mean CAR, 2.43%, giving ratios of 2.12 and 2.30. This agreement supports the scale of measured later purchases on the overlap. It does not fill missing early reports or establish that either channel captures every relevant investor.

A fixed fund-universe check further separates some coverage change from changes in event composition. Only three funds satisfy the selected mandate and report-availability requirements at every later-decade review; none do so across the whole 2007–2026 period. Within each block, using the same events for the fixed and broad fund measures, fixed-universe addition demand falls from approximately 0.49% to 0.38% between the two later blocks, compared with 1.11% to 0.97% in the broad measure. The smaller denominator describes a narrower investor population, not a correction that recovers total demand. Appendix C gives the definition and results.

The holdings evidence thus supports a measured trading association and a useful recent cross-source comparison. It does not establish a precise long-run elasticity trend. The difference from the anchor's US argument is substantive: here, the historical denominator is too coverage-sensitive to carry a strong claim about how efficiently the market absorbs a fixed amount of demand.

4.5 Trading concentration and liquidity

The return results do not mean that portfolio adjustment has become unimportant. The final trading day before effectiveness accounts for 9.44% of the 45-day volume window for early additions and 15.24% for latest-period additions. The increase is 5.79 percentage points, with a 95% interval of [3.30, 8.84]. Deletions show a similar increase, from 8.75% to 14.52%, a difference of 5.76 points with an interval of [3.26, 8.38].

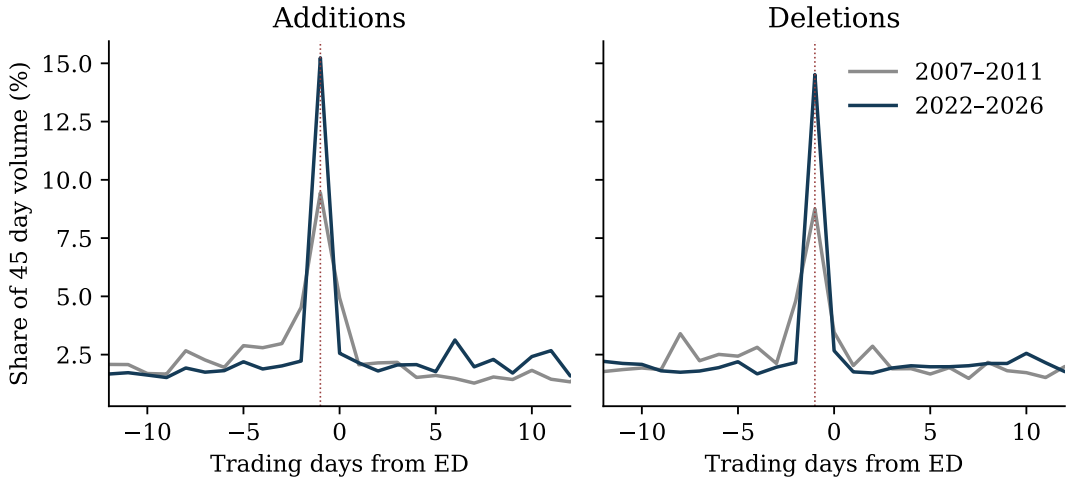


Figure 3: Trading concentrates immediately before effectiveness. Daily volume as a percentage of each event's total over ED-22 through ED+22; the graph displays ED-12 through ED+12. The dotted line marks ED-1, the final trading day before the new portfolio takes effect. Missing volume is set to zero in this measure. Daily observations do not separate auction and continuous trading.

More concentrated trading can coexist with a small implementation-window return. The latest addition implementation CAR is -0.09%, whereas the announcement CAR is 1.20%. Investors may anticipate demand or supply liquidity when funds rebalance; prices may also respond to information at other points in the interval. The daily data do not identify which explanation dominates. What they do establish is that the distribution of measured trading around implementation changes even when the mean return near implementation is small.

The spread series supplies context. The value-weighted closing quoted spread among OMXSB members falls from 0.270% in 2007 to 0.079% in 2025; the partial 2026 value is 0.087%. This is consistent with lower quoted trading frictions. It is a different object from the effective spreads and implementation costs examined in the US paper. A closing quote does not measure market depth or the price impact of a large predictable order, so the spread decline cannot establish that improved liquidity caused the return pattern. The annual series is retained in Appendix F.

5 Discussion and extensions

5.1 What the Swedish replication adds

The Swedish results qualify a simple disappearing-effect narrative in three ways. First, the later-decade addition return is positive under several separately reported specifications. Second, the early-to-late decline is not a steady progression, and its magnitude depends on which part of the reconstitution episode is measured. Third, smaller implementation-window returns coexist with more concentrated trading immediately before effectiveness. The contribution is the joint evidence on returns, measurement and trading in a scheduled-review setting, rather than a claim that a single coefficient overturns the US findings.

This distinction also clarifies the comparison with Greenwood and Sammon. Their question links a changing price response to index demand and market capacity to absorb it. The Swedish data allow a related investigation, but neither the country nor the period is held constant against the US. The shortened announcement lead time is an additional institutional feature that must enter the interpretation of the Swedish full-window series. A return over a shorter implementation interval is not automatically a weaker response to the announcement itself.

The fixed-window calculation makes this point concrete. The latest-minus-earliest addition contrast is approximately -7.03 percentage points for the full episode and +0.62 points for the common first seven daily returns. The remaining interval contributes -7.64 points to the difference. The initial seven-day contrast is imprecise, with a 95% analytical interval of [-1.80, 3.03]. We therefore do not establish equal announcement responses or a rise over time. We establish where the observed sample difference is located. Different securities, durations and intervening news remain possible explanations for the remaining return.

Measured fund demand gives a second substantive limit to the comparison. Recent matched data from FI and Lipper produce ownership changes of similar scale, which supports using the later sample to describe trading. The early multiplier is based on a small and selectively retained sample, however. It cannot carry a precise claim that a unit of index demand now moves prices a specified fraction as much as it did early in the sample. A convincing historical elasticity comparison would require comparable investor coverage and corporate-action treatment, not merely a more elaborate regression.

5.2 How much of the return remains after implementation?

Temporary trading pressure is one possible explanation for index-event returns. To describe their subsequent path, we extend the horizon through ED+20 and retain exactly the same events for every plotted day. The return before implementation runs from AD-1 through ED-1; the after-implementation component runs from ED through ED+20. Their sum includes every day once. This is a separate exploratory extension: Greenwood and Sammon's longer demand-ratio window instead starts earlier, at AD-20, and ends at ED+1.

For later-decade additions, the pre-implementation mean is 2.43% and the subsequent component is 0.87%, giving 3.30% cumulatively. The cumulative analytical interval is

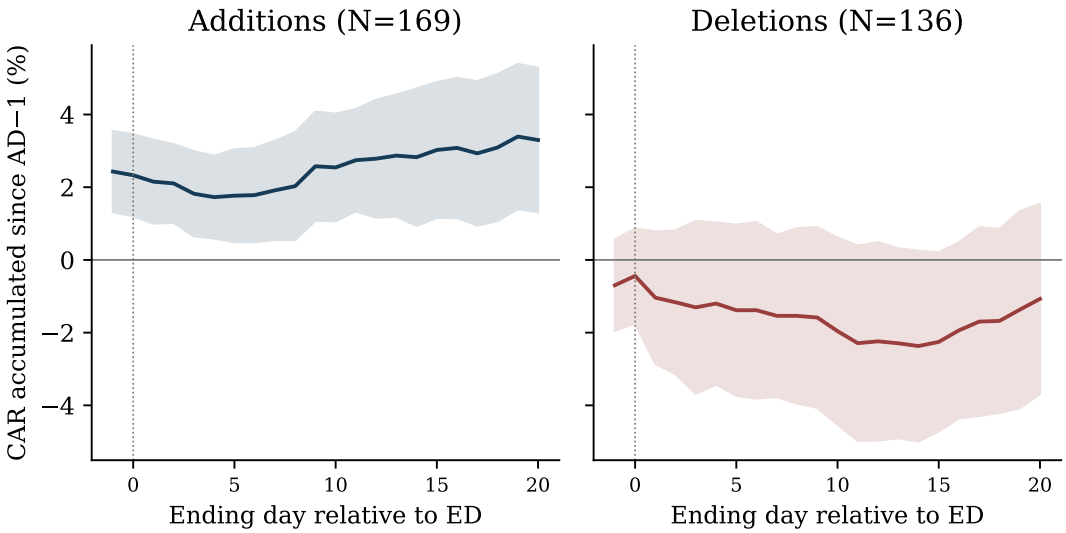


Figure 4: Cumulative performance through twenty days after implementation. The cohort is fixed separately for additions and deletions in 2017–2026. At every point, the cumulative return starts at AD-1; the horizontal axis gives its ending day relative to ED. The first plotted point is therefore the return already accumulated through ED-1, not a zero normalization. Shading gives pointwise 95% analytical intervals clustered by review, not a simultaneous confidence band. Counts are 169 additions and 136 deletions.

Table 10: Returns before and after implementation on a fixed cohort

Type	Period	Events	Before	After	Cumulative	95% interval
Addition	2017–2026	169	2.43	0.87	3.30	[1.31, 5.28]
Addition	2022–2026	87	1.77	-0.06	1.71	[0.72, 2.69]
Deletion	2017–2026	136	-0.70	-0.38	-1.07	[-3.68, 1.54]
Deletion	2022–2026	75	-0.93	-0.95	-1.88	[-5.72, 1.96]

Returns in percent. Before is AD-1 through ED-1; After is ED through ED+20; Cumulative is AD-1 through ED+20. Each row uses the same events throughout. Intervals are analytical CR1 with review clustering for the cumulative mean. Separate wild-bootstrap tests and post-result exclusions are in the appendix. The horizon is finite and the analysis exploratory.

[1.31%, 5.28%]. The after-implementation component's interval is [-0.78%, 2.51%], so the result does not establish a separately positive drift after the index change. It shows that the measured cumulative average has not returned to zero by the selected horizon.

The latest block illustrates why a longer horizon must not be selected simply because it produces a smaller p-value. Its cumulative estimate is 1.71%, with an interval of [0.72%, 2.69%], but falls to 1.25% without Sivers and 1.21% without MSCI overlaps; both latter intervals include zero. The pooled later-decade cumulative mean remains positive in those separate checks. The exclusions were examined after the extension's results were observed. All are reported as exploratory sensitivities, not independent confirmations.

Deletion cumulative returns are less precisely resolved: -1.07% in 2017–2026 and -1.88% in the latest block, with both intervals spanning zero. Neither side of the exercise establishes permanent price impact. Twenty trading days remain a finite horizon, and selection, subsequent news and benchmark error can affect cumulative performance throughout it.

5.3 A comparison within the same firm

Swedish firms with multiple listed share classes offer an additional comparison. Both classes are exposed to much of the same firm information, while only one may be involved in the index event. If class-specific demand is important, their relative performance is informative. This design improves on a comparison with an unrelated firm in one respect, but it does not make the other class untreated: substitution, arbitrage and shared investor attention can transmit the event to it.

Table 11: The indexed and sister share classes on the same events

Events	Count	Firms	Index class	Sister class	Difference
Addition	19	10	2.13	2.48	-0.34
Deletion	20	11	4.17	3.20	0.97
Combined	39	13	3.18	2.85	0.33

Sign-adjusted log market-excess returns in percent for AD-1 through ED+1. Deletion signs are reversed. Each difference subtracts the sister-class mean from the index-class mean on identical events. The combined sample contains 39 events from 13 distinct firms; firm counts by event type overlap. The combined relative-return placebo p-value is 0.407. These selected log-return observations cannot be subtracted from the main arithmetic-CAR sample to identify a causal component.

Table 11 reports both legs of the comparison on identical events. Across 39 events from 13 firms, the sign-adjusted log market-excess return is 3.18% for the index class and 2.85% for the sister class, leaving a relative return of 0.33 percentage points. The event-weighted placebo p-value for the relative return is 0.407. For the 19 additions alone, the corresponding means are 2.13% and 2.48%, giving a relative return of -0.34 points. These observations are a selected subset of the full sample and use log relative returns, so they cannot be subtracted from the 2.15% main estimate as a causal decomposition.

The small relative estimate does not establish that index demand has no effect. Some of the difference may be transmitted to the sister class; some firms occur repeatedly; and

thin trading or changes in return sources affect the measurement. A source-consistent subset contains 29 events from nine firms and has a relative mean near 0.04 points. The placebo construction does not jointly preserve all firm and review dependence. This extension therefore indicates that the available within-firm sample does not precisely isolate a class-specific response, while keeping both stock legs visible rather than treating the comparison as an unexplained null.

A separate size-screened comparison provides a complementary check. After removing the one-day overlap between its pre-return control and outcome, the later addition coefficient is 1.65 percentage points, with a 95% analytical interval of [0.48, 2.83]. Restricting comparisons to the main market gives 1.49 points, with an interval of [0.24, 2.74]. The correction preserves the sample and other controls. These remain conditional differences rather than threshold-based causal estimates; Appendix E reports the full before-and-after results.

5.4 Limits and implications

The main unresolved identification issue is that index membership is related to firm characteristics and prior performance. Market adjustment, composition controls and comparison securities do not remove every reason that selected stocks may have different returns. The study consequently establishes associations around observed changes rather than the causal effect of assigning a stock to OMXSB.

Measurement is a second constraint. Some historical notices still require field-by-field reconciliation; return availability excludes part of the source inventory; and absent holdings rows do not independently prove zero positions. The shares-outstanding denominator is available in the vendor panel on ED for every retained complete holdings event, but that does not establish the freshness of the underlying issuer filing. The adjustment screen reduces early coverage particularly heavily. These issues matter most for long-run demand comparisons and are recorded in Appendix C.

Statistical precision is a third constraint. Many securities share the same review, the latest period has only nine review clusters, and the full research record contains multiple outcomes and specifications. The uncertainty checks prevent a point estimate from being treated as more informative than it is; they are not a separate economic explanation for the result. Some sparse auxiliary bootstrap intervals also vary across numerical environments. The main conclusions do not rely on those unstable endpoints.

For investors, the evidence describes the timing and magnitude of observed returns; it is not a tested net trading strategy. The estimates do not deduct transaction costs, impose realistic execution rules or guarantee that membership changes were predictable. For research on indexing, the central implication is narrower: full-window returns, immediate announcement responses and implementation trading should be distinguished when the event calendar changes. In this sample, the case for a completely vanished Swedish addition association is weak, while the case for a precisely identified historical elasticity decline is weaker still.

6 Conclusion

This thesis replicates and extends Greenwood and Sammon's investigation of the index effect using OMX Stockholm Benchmark changes from July 2007 through June 2026. The complete return sample contains 256 additions and 215 deletions. Addition CAR falls from 8.52% in the earliest block to 1.50% in the latest, but the intervening periods do not form a steady decline. Across 2017–2026, the mean remains 2.15%, with a 95% interval of [1.01%, 3.39%]. The latest-period interval is too wide to establish that the return has become economically negligible.

The Swedish calendar adds an important qualification. Announcements occur much closer to implementation recently, and the first-to-last decline is concentrated after a common first seven daily returns. The initial seven-day means are 1.19% and 1.80%, with an imprecise difference. This does not show that shorter lead times caused the decline; it shows why a declining full-event return should not automatically be described as a declining announcement response.

Benchmark-linked holdings change around the events, and trading becomes more concentrated immediately before effectiveness. A recent FI comparison supports the scale of observed fund purchases. Historical coverage and adjustment restrictions nevertheless make the early return-to-demand ratio too fragile for a precise elasticity comparison. Within-firm share classes do not isolate a well-resolved class-specific response, while fixed-cohort returns through ED+20 retain a positive later-decade addition association.

The resulting conclusion is substantive but bounded: positive later Swedish addition returns coexist with concentrated implementation trading, and their apparent evolution depends on timing and measurement. The data establish neither complete disappearance nor a permanent causal price effect of index membership. The economic distinction is between returns accumulated over the full episode, the response during a fixed initial interval, and trading at implementation. In Sweden, these do not evolve as one interchangeable measure.

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A Replication details and source versions

This study replicates and extends the empirical framework of Greenwood and Sammon (2025) in Sweden. The detailed comparison uses their March 2024 author manuscript, deposited in Harvard DASH under the 2025 publication record (Greenwood and Sammon, 2024). That manuscript is not authenticated as the final published text: its abstract reports a recent deletion return of 0.1%, while its body retains an older estimate of −0.6%. Published headline comparisons therefore use the journal's abstract; the method references below identify the March 2024 version explicitly. The published full text and official replication files were not accessible for this review, leaving program-level equivalence unresolved.

The author manuscript's Equation (2) defines excess cumulative stock returns relative to the S&P 500 over announcement, implementation and announcement-to-implementation windows. Its Equation (1) motivates a price multiplier M applied to measured demand D. The Swedish analysis preserves these organizing questions, replacing the benchmark with OMXSBGI. It adapts the event screens, investor classifications, reporting intervals and inference to the available Swedish data. In particular, review clustering differs from the White standard errors in the original main tables; many Swedish securities change membership simultaneously.

The US tracker classification uses the average annual share of index trades and a 50% threshold. Swedish production estimates instead use dated fund mandates. The US observations straddle quarterly reports, while Swedish snapshots are sought before AD and after ED. The original acquisition and listing screens also differ from the Swedish offer/delisting exclusions. These departures preclude describing the exercise as an exact reproduction. Likewise, MSCI overlap is a concurrent-event sensitivity, not a reconstruction of the original S&P index-family migration mechanism.

The four composition controls retain the original categories: abnormal turnover, relative size, residual variance and analyst coverage. Their construction is adapted: the Swedish monthly forecast count need not equal the original unique-analyst count at the latest earnings announcement. Demand also uses ED shares outstanding rather than the original post-event quarterly denominator. The multiplier with controls scales period indicators by period-average demand; it is not an event-level regression of returns on individual fund purchases.

The pre-announcement exercise follows the original paper's anticipation question, but positive prior returns can also reflect selection on performance. The longer demand-ratio window is AD−20 through ED+1, as in the author manuscript's Tables 5–6. It should be distinguished from the separate Swedish extension through ED+20.

Table A1: The earlier-start demand-ratio window

Type	Period	Short N	Short M	Long N	Long M
Addition	2007–11	8	57.87	8	60.83
Addition	2012–16	27	-3.76	27	-1.82
Addition	2017–21	58	2.97	58	4.24
Addition	2022–26	78	2.35	75	6.66
Deletion	2007–11	7	-0.72	7	12.98
Deletion	2012–16	25	4.36	25	8.49
Deletion	2017–21	40	0.20	40	-0.57
Deletion	2022–26	68	1.41	67	2.44

Short is AD-1 to ED+1; Long is AD-20 to ED+1, the earlier-start counterpart in the anchor study. Each ratio uses identical events in its own numerator and denominator, but the short and long samples can differ. These are descriptive point estimates; the table does not claim well-identified early multipliers. The ED+20 analysis is a separate extension.

B Return conventions and source checks

The observation is a security–review event; an issuer can recur or contribute different share classes. Dates are mapped to the observed trading calendar. Every stated CAR window includes the daily returns at both endpoints. Thus AD−1 is the first included daily return, not the closing-price baseline for the subsequent investment. CAR sums daily percentage stock returns less daily OMXSBGI returns. Stock total returns are used where available, with closing-price returns as a fallback. This convention does not automatically preserve dividends in every fallback observation.

The source manuscript calls its inputs cumulative returns but does not settle the daily aggregation or endpoint conventions sufficiently to authenticate an exact match. Table A2 therefore compares arithmetic sums with compounded stock returns minus compounded benchmark returns on identical observations and included days. For 2017–2026 additions, the means are 2.15% and 2.18%; in 2022–2026 they are 1.50% and 1.45%. Interpreting AD−1 as a baseline close instead changes the later-decade mean to approximately 2.75%. That is a change of included returns, not a compounding result.

The announcement-to-effective lag ranges from six to eighteen trading days. It is generally fourteen to eighteen days during 2010–2019 and six to eight days during 2022–2026. Consequently, the total event window changes duration. The common AD−1 through AD+5 sensitivity ends before ED in every review; its complement begins at AD+6. Their sum exactly recovers the total CAR on the same cohort. The first-to-last addition decline is concentrated in this remainder, whose duration and composition also vary. This accounting does not show that shorter notice caused lower returns.

External research recovered accessible notices or original attachments for 35 of 39 reviews, sometimes through republication; three further reviews were visible only in indexed text. The 2008 second-half notice remained unresolved. A December 2008 correction also prints an inconsistent effective year. Document retrieval is not complete field-level reconciliation, and neither a trading-volume peak nor the label “effective date” independently authenticates open-versus-close implementation conventions.

Table A2: Return definitions on matched events

Type	Period	Events	Daily sum	Compounded	Base at AD-1
Addition	2007–2011	33	8.52	8.60	9.27
Addition	2012–2016	54	0.25	0.38	0.42
Addition	2017–2021	82	2.85	2.96	4.20
Addition	2022–2026	87	1.50	1.45	1.38
Addition	2017–2026	169	2.15	2.18	2.75
Deletion	2007–2011	33	-6.16	-6.05	-6.61
Deletion	2012–2016	46	-2.47	-2.38	-2.62
Deletion	2017–2021	61	-0.43	-0.53	-0.37
Deletion	2022–2026	75	-1.53	-1.63	-1.87
Deletion	2017–2026	136	-1.04	-1.14	-1.20

Daily sum is the primary arithmetic CAR. Compounded is the stock buy-and-hold return minus the benchmark buy-and-hold return using the same included daily returns, AD-1 through ED+1. Base at AD-1 instead starts from the AD-1 close and includes daily returns AD through ED+1. Both latter columns compound stock and benchmark separately. These are sensitivity checks, not proof of the original US code's convention.

C Holdings measurement and coverage

Group C combines strict followers, benchmark variants and funds with an OMXSB-linked component. It measures covered positions rather than all passive capital. The numerator reconstructs to numerical precision for all 311 complete primary holdings events. Available timestamps satisfy the stated pre-AD and post-ED limits, but this verifies recorded dates rather than the completeness of the underlying reports. Forty fund-event records use a pre-position date different from the latest fund report identified by the availability check.

An absent position is treated as zero when other reporting establishes fund availability. Among addition records, 1,630 of 1,739 contributing fund-event observations have no observed pre-position row. A zero is plausible for a newly included security, but a security-query extract cannot independently distinguish it from an omitted response. Original reports or vendor completeness information are required to resolve that distinction.

The adjustment exclusion compares a market-capitalization/share-count/adjusted-price proxy at ED with its value at the security's final available price date. A missing ratio or a ratio outside $[1/1.4, 1.4]$ triggers exclusion. It is therefore a share-adjustment consistency screen, not simply a record of splits during the event window. Later corporate actions can affect early-event eligibility. This safeguard may be needed because adjusted fund shares are divided by historical shares outstanding; removing it without reconciling vendor conventions would not be a justified correction.

All 468 stored share denominators and adjustment ratios reconstruct. For the 311 retained holdings observations, the denominator exists on ED in the daily panel, although daily availability does not prove that the underlying issuer disclosure is fresh. The screen removes 26 of 37 early addition candidates, compared with nine of 87 in the latest block, contributing to unequal historical coverage.

A fixed-universe sensitivity intersects verified mandate status and reporting availability across every review, without selecting on purchases or returns. No fund survives all 39

Table A3: Fund definitions on the broad-measure event sample

Type	Period	Events	Strict	+ variants	Broad	Mixed count
Addition	2007–11	8	0.10	0.15	0.15	0.0
Addition	2012–16	27	0.37	0.49	0.52	0.4
Addition	2017–21	58	0.92	1.02	1.11	1.0
Addition	2022–26	78	0.41	0.78	0.97	8.8
Deletion	2007–11	7	0.20	0.26	0.26	0.0
Deletion	2012–16	25	0.31	0.46	0.48	0.1
Deletion	2017–21	40	0.83	0.92	0.99	1.0
Deletion	2022–26	68	0.52	0.93	1.18	9.4

Demand is percent of shares outstanding, sign-adjusted for deletions. All three demand definitions in a row use the same retained event sample. Strict comprises exact benchmark followers; the next column adds related benchmark variants; Broad also includes the benchmark-linked mixed-fund category. Mixed count is the mean number of reporting mixed funds per event, not their ownership percentage. These data do not measure the complete investor universe.

reviews. Three survive the 19 later reviews: Swedbank Robur Access Mix, Swedbank Robur Access Sverige and iShares OMX Stockholm Capped UCITS ETF. Within each block, on the same additions as the broad measure, their mean purchases fall from 0.488% to 0.378% between the last two blocks, compared with 1.107% and 0.973% for broad C. The narrower measure retains the same missing-position assumptions and omits relevant investors. It supports neither continuously increasing purchases nor a corrected estimate of total index demand.

D Statistical inference and sensitivity

For the event-weighted mean $\widehat{\mu} = N^{-1} \sum_i y_i$, review-clustered CR1 variance is

$$\widehat{V}(\widehat{\mu}) = \frac{G}{G-1} \frac{1}{N^2} \sum_{g=1}^{G} \left[\sum_{i \in g} (y_i - \widehat{\mu}) \right]^2, \tag{4}$$

where G is the number of reviews. The intercept-only regression's remaining degrees-of-freedom correction equals one. Joint covariance of mean signed returns and demand uses the corresponding two-dimensional review scores.

For the ratio $\widehat{M} = \overline{Y}/\overline{D}$, a candidate value m belongs to the Fieller set if

$$(\overline{Y} - m\overline{D})^2 \leq t^2_{G-1,0.975} \left[\widehat{V}_{YY} - 2m\widehat{V}_{YD} + m^2 \widehat{V}_{DD} \right]. \tag{5}$$

This accounts for covariance between numerator and denominator (Fieller, 1954). Sets can be bounded, unbounded or all real values; weak denominator precision must not be concealed by discarding inconvenient outcomes. With few reviews, the approximation does not guarantee exact coverage. The early addition set spans $[-9.55, 471.88]$, while the earliest deletion set contains every real value. These estimates cannot establish a precise historical change in structural elasticity.

Table A4: Returns scaled by observed benchmark linked fund purchases

Type	Period	N	G	$\bar{Y}$	$\bar{D}_C$	$\widehat{M}$	Fieller set
Addition	2007–11	8	5	8.78	0.15	57.87	[-9.55, 471.88]
Addition	2012–16	27	8	-1.94	0.52	-3.76	[-15.09, 4.24]
Addition	2017–21	58	10	3.29	1.11	2.97	[-0.14, 6.30]
Addition	2022–26	78	9	2.28	0.97	2.35	[-0.21, 4.73]
Addition	2017–26	136	19	2.71	1.03	2.63	[0.91, 4.37]
Deletion	2007–11	7	3	-0.19	0.26	-0.72	All real values
Deletion	2012–16	25	7	2.07	0.48	4.36	[-17.73, 30.17]
Deletion	2017–21	40	10	0.20	0.99	0.20	[-4.05, 4.89]
Deletion	2022–26	68	9	1.66	1.18	1.41	[-0.87, 2.83]
Deletion	2017–26	108	19	1.12	1.11	1.01	[-0.88, 2.61]

$Y = s_i CAR_i[AD-1, ED+1]$ and D_C is the sign-adjusted change in covered benchmark-linked funds' holdings as a percentage of shares outstanding; $s_i = 1$ for additions and -1 for deletions. Positive values therefore indicate movement in the index-directed direction for both event types. $\widehat{M} = \bar{Y}/\bar{D}_C$ uses the same adjustment-screened complete-case sample for both means. Group C includes strict followers, benchmark variants, and benchmark-linked mixed-fund holdings. The last column is a 95% Fieller confidence set using the joint CR1 covariance of both means and t_{G-1}, an asymptotic sensitivity measure rather than exact small-sample inference. It differs from the percentile bootstrap. Source: Authors' calculations; exact inputs are listed in the replication package.

D.1 Weighting and dependence

Table A5: Weighting and dependence sensitivity

Type	Period	Event	Review	Leave one out	CR1	CV3	Two way
Addition	2017–2026	2.15	2.50	[1.87, 2.47]	0.55	0.55	0.47
Addition	2022–2026	1.50	1.50	[1.02, 2.01]	0.74	0.77	0.71
Deletion	2017–2026	-1.04	-0.52	[-1.45, -0.50]	0.87	0.88	0.87
Deletion	2022–2026	-1.53	-1.32	[-2.06, -0.58]	1.12	1.17	1.11

CAR $[AD-1, ED+1]$ in percent. Review mean gives each review equal weight. Leave one out is the range of event-weighted estimates after removing one review at a time, not a confidence interval. CR1, CV3 and Two way report standard errors. CV3 is the cluster jackknife centered on the full-sample estimate. Two way combines review and security clustering with subtraction of the review-security intersection. It does not solve dependence between different securities across different reviews. Source: Authors' calculations; exact inputs are listed in the replication package.

Equal-event and equal-review means answer different averaging questions. Leave-one-review-out ranges describe influence rather than confidence coverage. The jackknife and review-by-security covariance checks address additional dependence assumptions; two-way clustering does not capture every correlation across different securities and reviews (Petersen, 2009; Cameron et al., 2011; MacKinnon et al., 2023).

Table A6: Changes in the event window return

Type	Contrast	Estimate	95% interval	p
Addition	Latest minus earliest	-7.03	[-13.10, -1.10]	0.029
Addition	Latest minus preceding	-1.35	[-3.88, 0.98]	0.250
Addition	Later minus earlier decade	-1.24	[-5.26, 2.54]	0.502
Addition	Linear trend per year	-0.28	[-0.64, 0.08]	0.117
Deletion	Latest minus earliest	4.62	[0.25, 9.81]	0.039
Deletion	Latest minus preceding	-1.10	[-5.11, 2.70]	0.564
Deletion	Later minus earlier decade	2.97	[-2.94, 8.60]	0.328
Deletion	Linear trend per year	0.19	[-0.29, 0.64]	0.400

The outcome is CAR $[AD - 1, ED + 1]$. Differences are percentage points; the linear trend is percentage points per calendar year. Earliest, preceding, and latest periods are 2007–2011, 2017–2021, and 2022–2026. Decades are 2007–2016 and 2017–2026. Restricted wild bootstrap inference clusters by review cycle, using 9,999 draws. A difference of significance levels across periods is not a test of a difference. Source: Authors' calculations; exact inputs are listed in the replication package.

Table A7: Multiplicity in a declared ten test audit family

Type	Test	Raw p	Holm p
Addition	2007–2011	0.027	0.214
Addition	2012–2016	0.908	1.000
Addition	2017–2021	0.005	0.046
Addition	2022–2026	0.062	0.310
Addition	2022–26 minus 2007–11	0.029	0.214
Deletion	2007–2011	0.007	0.059
Deletion	2012–2016	0.522	1.000
Deletion	2017–2021	0.759	1.000
Deletion	2022–2026	0.279	1.000
Deletion	2022–26 minus 2007–11	0.039	0.235

The audit family contains eight five-year total-window means and two latest-minus-earliest contrasts. It was declared after the results were known. Holm adjustment controls familywise error for this family under valid marginal tests; it cannot correct unrecorded historical specification search. It does not turn the thesis into a preregistered study. Source: Authors' calculations; exact inputs are listed in the replication package.

D.2 Multiplicity, equivalence and precision

Only the 2017–2021 addition mean remains below 0.05 after Holm adjustment within the declared ten-test family (Holm, 1979). That family was specified after inherited results were known and excludes many historical specifications. It is a limited sensitivity, not a retrospective confirmatory design.

The archived minimum-detectable-effect approximation is

$$MDE \approx \left(t_{G-1,0.975} + t_{G-1,0.80}\right) \widehat{SE}. \tag{6}$$

It describes the scale associated with an approximate two-sided 5% test at 80% power, not exact bootstrap power. The latest addition MDE is 2.35 percentage points. Its analytical 90% interval, $[0.13, 2.87]$, does not establish equivalence within the illustrative ± 1 percentage-point margin; neither does the deletion interval $[-3.62, 0.55]$. The margin was chosen after existing results were known.

D.3 Timing and longer-horizon checks

Table A8: First-to-last changes in the timing decomposition

Type	Component	Difference	95% CR1 interval	p
Addition	Full window	-7.03	[-11.97, -2.08]	0.027
Addition	Initial seven days	0.62	[-1.80, 3.03]	0.625
Addition	Remaining interval	-7.64	[-11.91, -3.37]	0.009
Deletion	Full window	4.62	[0.38, 8.87]	0.040
Deletion	Initial seven days	2.78	[-0.10, 5.66]	0.088
Deletion	Remaining interval	1.84	[-2.35, 6.04]	0.404

Differences are 2022–2026 minus 2007–2011, in percentage points, from a four-period indicator regression. Intervals are analytical CR1 using review clusters; two-sided tests use 49,999 restricted wild draws. These are pointwise exploratory tests, not adjusted for all prior specification choices. The component contrasts add to the full contrast.

The fixed-window decomposition and ED+20 extension are exploratory. New analytical intervals use CR1 and t_{G-1}, while separately reported restricted wild-bootstrap tests use 49,999 draws and finite-simulation correction. They are not inverted versions of those intervals. News and concurrent-index exclusions change the sample and remain incomplete. A positive cumulative return through ED+20 neither establishes permanent impact nor proves positive post-effective drift. All periods and both event types are retained in the supporting outputs.

Table A9: Exclusions in the exploratory cumulative return extension

Period	Sample	N	CAR	CR1 interval	p_{WCR}
2017–2026	full	169	3.30	[1.31, 5.28]	< 0.001
2017–2026	without Sivers	168	3.07	[0.96, 5.18]	0.003
2017–2026	without MSCI overlap	157	2.51	[0.60, 4.42]	0.008
2017–2026	without screened company news	163	3.14	[1.13, 5.15]	< 0.001
2022–2026	full	87	1.71	[0.72, 2.69]	0.010
2022–2026	without Sivers	86	1.25	[-0.14, 2.63]	0.087
2022–2026	without MSCI overlap	82	1.21	[-0.25, 2.67]	0.114
2022–2026	without screened company news	84	1.79	[1.04, 2.53]	0.003

CAR $[AD-1, ED+20]$ in percent, constructed on complete non-overlapping before/after cohorts. Each exclusion is separate. These follow-up diagnostics were added after inspecting the new cumulative estimates, and inherit the limitations of the original screens. Analytical intervals and 49,999-draw wild-bootstrap p-values have the same definitions as Table 10. Source: Authors' calculations; exact inputs are listed in the replication package.

E Repaired comparison-security estimates

Table A10: Removing overlap from the comparison control

Type	Period	Events	Controls	Before	After	After 95% interval
Addition	2007–2016	78	787	3.90	4.01	[0.14, 7.89]
Addition	2017–2026	159	1609	1.53	1.65	[0.48, 2.83]
Deletion	2007–2016	68	415	-3.24	-3.42	[-8.20, 1.36]
Deletion	2017–2026	130	833	-1.40	-1.57	[-3.04, -0.10]

Coefficients are percentage points on the event indicator in the same review-fixed-effect, log-size and pre-return regression. Before includes AD-1 in the pre-return control; After removes only that day. Outcomes, sample and other regressors are unchanged. Intervals are analytical CR1 clustered by review; separate 49,999-draw wild tests and the main-market restriction are provided in the machine-readable results. Size screens are not reconstructed Nasdaq assignment thresholds.

The inherited comparison regression controlled for returns from AD−100 through AD−1, sharing its final day with the outcome. The repair ends that control at AD−2. It preserves outcomes, security selection, review effects, capitalization controls and sample sizes. The later addition coefficient becomes 1.65 percentage points with all comparison securities and 1.49 with main-market comparisons, with restricted wild-bootstrap p-values of 0.011 and 0.029. Both use 159 additions over 19 reviews. Table intervals are analytical CR1 intervals, not bootstrap inversions.

The size-based comparison screen is not a Nasdaq assignment threshold. These estimates remain conditional return differences: neither removing the overlap nor adding review effects establishes comparable unobserved fundamentals. The full output retains early periods, deletion models and alternative exclusions.

F Additional evidence

The OMXS30 exercise contains 36 events across 15 reviews. Addition CAR averages 0.21%, with interval $[-2.59, 3.27]$; sign-adjusted deletion CAR averages 0.69%, with interval $[-3.35, 7.16]$. These estimates are too imprecise to establish disappearance or equality across indices with different firms and selection rules.

The issuance exercise measures shares outstanding from AD−5 to ED+60. Its continuous estimate is 0.57 percentage points, with interval $[-0.47, 1.65]$. Binary thresholds produce different nominal significance decisions. Forty-eight of 2,586 rows are flagged as split-like, while 23 changes exceeding 30 log percentage points remain. Without complete corporate-action reconciliation, these results do not establish an issuer response to index demand.

Closing quoted spreads measure displayed bid–ask differences, not transaction-level effective spreads or implementation shortfall. Annual return means reveal uneven historical outcomes but can rest on very few reviews. The figures are contextual diagnostics rather than independent identification tests.

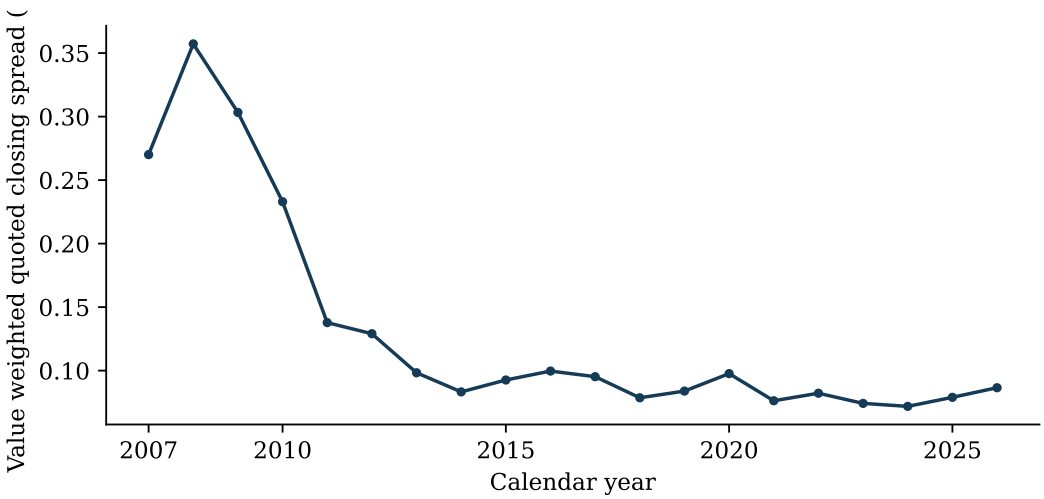


Figure A1: Annual closing quoted spreads among OMXSB members. These vendor quote measures differ from the effective-spread and execution-cost evidence in Greenwood and Sammon.

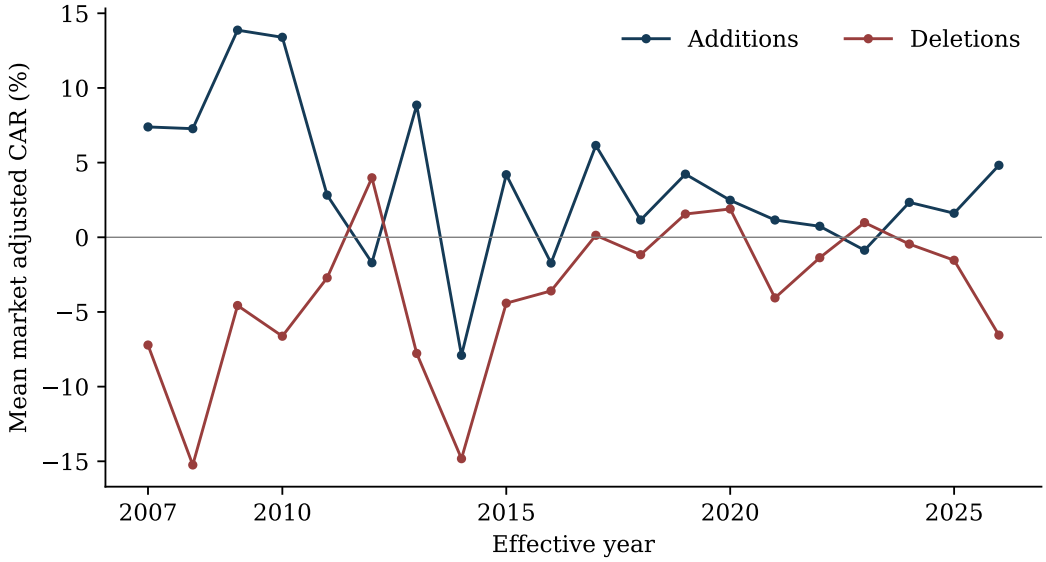


Figure A2: Annual event-return means. Annual averages describe the historical sample; sparse annual review counts limit their precision.

G Reproducibility and disclosure

The analysis is anchored to repository commit `d2d41666ed21`, with the complete identifier and input hashes in the supporting manifest. Independent reconstruction verifies the main return arithmetic and selected estimators. Rerunning 40 numerical outputs reproduces point estimates and counts; eight differ in simulated inference. Some sparse auxiliary ratio intervals are unstable, so conclusions do not rely on those endpoints. These checks verify specified computations, not every source observation or identifying assumption.

The support package separates frozen estimates from repairs and exploratory additions, and records aggregate results, code, references and unresolved source checks. Licensed security-level data and third-party full texts are not newly redistributed. Reproduction requires authorized access to the documented inputs.

The paper's structure follows the six-part sequence in the BE451 course material. Earlier SSE theses by Fager and Öhman Tunklint (2024), Holst and von der Esch (2023) and Domeij and Krieg (2021) informed the emphasis on a focused question, a close benchmark study and self-contained exhibits. These examples were examined by Adrien d'Avernas; they are not evidence of an examiner-specific template for Riccardo Sabbatucci. The supporting structure note records the comparison.

AI disclosure for this working document. OpenAI Codex assisted with retrieval, calculations, code review, exhibits and manuscript composition; this prose is AI-generated. The BE451 syllabus permits disclosed assistance but prohibits AI directly writing thesis text (Stockholm School of Economics, 2026). This document is a research aid, not a submission-compliant manuscript. The authors must write the submitted prose, verify the findings and disclose assistance accurately.